

# Scalability & Future Plan

|               |                |
|---------------|----------------|
| Date          | 15 March 2026  |
| Team ID       | XXXXXX         |
| Project Name  | PocketSmart AI |
| Maximum Marks | 1 Mark         |

The template focuses on larger user bases, increased data loads and future feature evolution. █filecite█turn15file11█L7-L24█

## Current System Limitations

| S.No | Limitation                                        | Impact                                                     | Priority to Address |
|------|---------------------------------------------------|------------------------------------------------------------|---------------------|
| 1    | AI/external API response latency and availability | Recommendation response time may vary.                     | High                |
| 2    | Limited initial recommendation data coverage      | Some product/service categories may have fewer options.    | Medium              |
| 3    | Prototype-scale deployment                        | Higher concurrent traffic may require scaling and caching. | High                |

## Scalability Plan

| S.No | Scalability Aspect | Current State                               | Proposed Upgrade / Solution                                                    |
|------|--------------------|---------------------------------------------|--------------------------------------------------------------------------------|
| 1    | User Load          | Prototype / initial deployment              | Stateless API services, horizontal scaling and load balancing.                 |
| 2    | Data Storage       | Small-scale structured data                 | Managed PostgreSQL with indexing, backups and read replicas when required.     |
| 3    | Performance        | AI/API calls can be latency-sensitive       | Caching, asynchronous processing, request timeouts and response optimization.  |
| 4    | Security           | Environment-based secret management planned | Secret manager, authentication/authorization, rate limiting and audit logging. |

## Future Roadmap

| Phase   | Planned Feature / Enhancement                                  | Target Timeline | Expected Impact                                            |
|---------|----------------------------------------------------------------|-----------------|------------------------------------------------------------|
| Phase 1 | Complete core recommendation prototype                         | Current phase   | Validate the main user journey and recommendation quality. |
| Phase 2 | Add broader product/service integrations                       | Next phase      | Increase recommendation coverage and usefulness.           |
| Phase 3 | Personalized profiles and recommendation history               | Future          | Improve personalization and repeat usage.                  |
| Phase 4 | Advanced optimization, analytics and scalable cloud deployment | Future          | Support larger user base and richer budget planning.       |